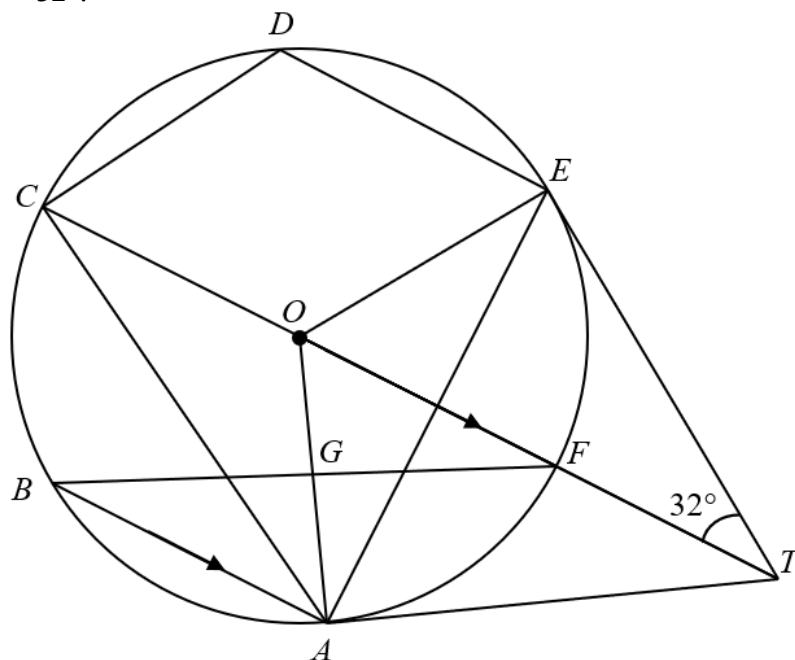


- 3** The diagram shows a circle $ABCDEF$, with centre O .
 TA and TE are tangents to the circle.
 BGF and COT are straight lines.
 OA intersects BF at G . CT is parallel to BA .
Angle $OTE = 32^\circ$.



- (a)** Identify the triangle that is congruent to triangle TOE and show that they are congruent.
Give a reason for each statement you make.

Answer

[3]

(b) Find, giving reasons for each step of your working,

(i) angle OFG ,

Answer angle OFG = _____ [3]

(ii) angle CDE .

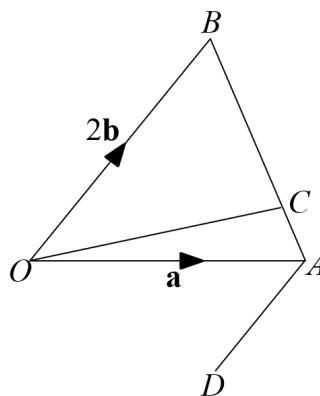
Answer angle CDE = _____ [3]

(c) Explain why points O, E, T and A can also be points on the circumference of another circle.

Answer _____

[1]

4



The position vectors of A and B , relative to O , are $\mathbf{a}$ and $2\mathbf{b}$ respectively.

$$\overrightarrow{BC} = \frac{3}{4} \overrightarrow{BA} \quad \text{and} \quad \overrightarrow{AD} = \frac{1}{2} \overrightarrow{BO}.$$

(a) Find, in terms of $\mathbf{a}$ and $\mathbf{b}$,

(i) $\overrightarrow{AB}$,

Answer $\overrightarrow{AB} =$ [1]

(ii) $\overrightarrow{OC}$.

Answer $\overrightarrow{OC} =$ [2]

(b) (i) P is a point outside of triangle ABO .

Find $\overrightarrow{OP}$, such that $\overrightarrow{BP} = 3\overrightarrow{OA}$.